

COMSATS University Islamabad (CUI), Islamabad Campus

Department of Computer Science

Internship Weekly Progress Report - Week 1 & Week 2

Student Name: [Your Name]

Registration Number: [Your CUI Registration Number]

Degree Program: BS Computer Science

Host Organization: Ezitech Engineering

Project Title: MediSight AI (Clinical Decision Support Platform)

Academic Supervisor: [Faculty Supervisor Name]

Site Supervisor: [Host/Site Supervisor Name]

Week 1 Progress Report

- **Reporting Period:** Week 1 (40 Hours)
- **Domain Focus:** Database Modeling, Monorepo Scaffolding & System Security (RLS)

1. Objective of the Week

The primary objective of Week 1 was to establish the structural and security foundation of the MediSight AI monorepo. This included designing the database schema, creating table relations for multi-modal patient records, establishing Row-Level Security (RLS) policies for HIPAA/GDPR compliance, and setting up the authentication flow for hospital staff (doctors, administrators, and assistants).

2. Nature of Tasks and Responsibilities Undertaken

As a Full-Stack Machine Learning Engineer Intern, my responsibilities during the first week included:

- **Monorepo Scaffolding:** Initializing the project workspace and separating concerns into ``/frontend`` (Next.js 14 Web Application), ``/ml-service`` (FastAPI Python backend), and ``/supabase`` (database migrations and serverless edge logic).

- **Database Schema Design:** Authoring custom SQL migration files to create 14 interconnected relational tables (``profiles``, ``hospitals``, ``patients``, ``vitals``, ``lab_results``, ``medications``, ``diagnoses``, ``predictions``, ``explanations``, ``alerts``, ``audit_logs``, ``doctor_patient_assignments``, ``ai_chat_sessions``, ``ai_messages``).

- **Row-Level Security (RLS) Implementation:** Writing custom SQL policy rules to restrict data access, ensuring that doctors can only view, create, or modify records of patients belonging to their assigned hospital departments.

- **Authentication Infrastructure:** Configuring Supabase Auth redirects, including multi-factor authentication (MFA) requirements for administrative and medical accounts, and password recovery redirection routes.

3. Technologies, Frameworks, and Tools Used

- **Frontend Core:** Next.js 14 (App Router), TypeScript, Tailwind CSS, shadcn/ui.
- **Backend Database:** PostgreSQL, Supabase Local Development Environment.
- **Version Control:** Git, GitHub for repository synchronization.
- **Environment Configuration:** Dotenv modules (``.env.example`` configurations).

4. Learning Experience (Knowledge & Skills Gained)

- **Technical Knowledge Gained:** Mastered the creation of granular security policies (RLS) at the database engine level. Understood how to map database structures to front-end sessions using modern Server Components.

- **Professional Skills Learned:** Gained insights into HIPAA compliant data engineering and the role of identity management in clinical settings.

5. Challenges Faced and Solutions Applied

- **Challenge:** Encountering recursive dependency conflicts when trying to apply database migrations containing self-referencing foreign keys (e.g., matching doctors with profiles and clinical departments).

- **Solution:** Re-ordered the migration execution sequence, separating the creation of core profiles from relational assignment tables, and utilized independent reference constraints.

6. Connection to Academic Preparation

This week's assignments directly bridged the concepts learned in **CSC371 (Database Systems)** and **CSC325 (Software Engineering)**. Implementing relational constraints, index optimizations, and database policies allowed me to apply academic transactional theory to a real-world enterprise healthcare system.

Week 2 Progress Report

- **Reporting Period:** Week 2 (40 Hours)
- **Domain Focus:** Synthetic Ingestion, Feature Engineering & ML Pipeline Development

1. Objective of the Week

The primary objective of Week 2 was to implement the machine learning pipeline. This involved generating clinical synthetic datasets, designing a feature engineering pipeline to compute medical biomarkers, training multiple predictive models, selecting the best-performing production candidate, and integrating explainability parameters via SHAP.

2. Nature of Tasks and Responsibilities Undertaken

My tasks and responsibilities during the second week included:

- **Dataset Generation & DB Seeding:** Populating the Supabase database with 400 realistic, de-identified patient records across four key categories (Diabetes, Stroke, Heart Disease, Kidney Disease) and 2,100 compliant audit-logged events.

- **Feature Engineering Module Development:** Creating a reusable Python feature transformer (`feature_engineer.py`) that computes clinical indexes, including:

- WHO Body Mass Index (BMI).
- AHA Blood Pressure stages (Mean Arterial Pressure, BP flags).
- eGFR (Estimated Glomerular Filtration Rate) using Age, Sex, and Serum Creatinine values.

- Composite cardiovascular risk index.
- **ML Model Training & Experiment Tracking:** Training baseline Logistic Regression, Random Forest, XGBoost, and LightGBM models, utilizing MLflow to track parameters, training metrics, and artifact outputs.
- **Model Evaluation & Explainability:** Evaluating model AUC-ROC and Recall. Integrating **SHAP (SHapley Additive exPlanations)** to extract global feature contributions and local waterfall plots for clinical clarity.
- **FastAPI Prediction Microservice:** Building the prediction REST endpoints (`/predict``, `/explain``, `/retrain``, `/model-status``) to serve model predictions in real-time.

3. Technologies, Frameworks, and Tools Used

- **Programming Languages:** Python 3.10.
- **Libraries & Core Engines:** Scikit-learn, XGBoost, LightGBM, SHAP, MLflow, SQLAlchemy, Pandas, NumPy, FastAPI.
- **Testing Suite:** Pytest for feature calculation validation.

4. Learning Experience (Knowledge & Skills Gained)

- **Technical Knowledge Gained:** Deepened understanding of handling extreme class imbalance in medical data using class weighting (`class_weight='balanced'`). Mastered the use of local explainability frameworks (SHAP) to interpret black-box model decisions.
- **Professional Skills Learned:** Learned how to log model runs, registry schemas, and parameters systematically, which is crucial for MLOps tracking and audit trails.

5. Challenges Faced and Solutions Applied

- **Challenge:** High class imbalance in the training datasets caused models to show high accuracy but poor Recall, which could result in dangerous missed diagnoses in a clinical environment.
- **Solution:** Configured model hyperparameters with custom positive class weights (e.g., `scale_pos_weight`` in XGBoost and `class_weight='balanced'` in LightGBM) to prioritize Recall over Precision, achieving a validated AUC-ROC of **0.8307** and reducing False Negatives.

6. Connection to Academic Preparation

The work completed during this period drew heavily on **CSC483 (Machine Learning)** and **CSC312 (Artificial Intelligence)**. Applying feature scaling, hyperparameter tuning, model performance evaluation, and ensemble strategies turned theoretical learning into an active predictive service.